

hyperparameters are very specific to the type of machine learning model you are trying to optimize. The simplest models have no hyperparameters. Examples might make the concept clearer. Although model specific, here are a few general hyperparameters:

- Learning Rate
- Number of hidden layers
- The k in k-nearest neighbors

Now that we know the meaning of these terms, how will this knowledge help us?

Data split ratio

Datasets do not come in train, validation and test variants. You might be thinking how does one split the dataset to get optimal results. Splitting of data is dependent on two things :

- the quantity and variety of data,
- the model being trained.

A few models need a sizeable amount of data to train upon, so in this case, you would opt for a larger training sets. Models with a lesser number of hyperparameters would be easier to tune and validate, the size of the validation set can be reduced. On the other hand, if your model has a greater number of hyperparameters, a large validation set is required (there are other methods to tackle the problem like k fold cross-validation). The validation set can be totally ignored if the model lacks any hyperparameters. Train-test-validation split like most other things in machine learning depends on the model being used. There are a few guidelines which are widely accepted but there is no hard and fast rule to stick by.

Training

Now that we have a basic understanding of the dataset and its effect on the model, let us discuss how we can use that knowledge to our advantage.

To train an ML model, you need to specify the following:

- Input data source
- The type of model
- A target attribute which contains the desired output (in case of supervised learning)
- Data transformation instructions
- Training parameters
- Hyperparameters